Dylan James-Kavanaugh

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Summary

Team oriented robotics engineer with expertise in autonomous systems regarding state estimation, sensor fusion, and adaptive control. Proven experience developing and deploying navigation algorithms, deep learning perception modules, and validation frameworks in ROS from simulation to hardware testing. Committed to advancing autonomous capabilities through innovative control theory, robust system integration, and cross functional collaboration.

Experience

Robotics Relevant Experience

Research Engineer — *San Diego, CA*
DSIM Lab San Diego State — *Aug 2021 - May 2025*
- Led technical design and deployment of adaptive control architecture within ROS for a custom TurtleBot3 mobile robot, establishing system requirements and validation strategies across 6 autonomous navigation control approaches for GPS-denied environments
- Developed an Extremum Seeking Control (ESC) algorithm for autonomous signal source convergence, integrated with sensor fusion techniques combining IMU, acoustic sensors, photoresistors (light sensors), LiDAR, and odometry to enable robust state estimation in unknown environments improving power by reducing the consumption by 88% from past research
- Designed verification and validation frameworks within the Gazebo physics simulator to ensure efficient performance of path planning algorithms prior to hardware deployment in a 25 m^2 controlled test environment
- Optimized the control algorithm design from previous research by over 1000%, resulting in improved robot navigation and convergence to a scalar signal
- Served as the technical lead for autonomous mobile robot design, overseeing stakeholder requirements, implementing unknown path planning algorithms, and coordinating contributions from multiple researchers using GitLab for version control and project management with over 6,000+ lines of code in Python and C++
- Collaborated with 6 person research team, on the development of intelligent autonomous robots, supporting efforts to advance novel control theory approaches for intelligent systems

Teaching Assistant — *San Diego, CA*
San Diego State University — *Aug 2023 - Dec 2024*
- Mentored and assisted over 240 students by evaluating student performance on assignments and exams for System Modeling and Robotics Motion Control, ensuring fair assessment and feedback to promote student learning
- Collaborated with course instructor to develop curriculum and exercises that align with robotics, contributing to a dynamic and engaging learning environment

Course Designer (Deep Learning) — *Barcelona, Spain*
The Construct: Robotics Careers — *Jan 2024 - Nov 2024*
- Designed and optimized Gazebo simulation environment to ensure alignment of deep learning concepts with students' learning curves and to improve engagement with robotics coursework, reaching thousands of students worldwide
- Developed advanced computer vision modules using PyTorch and OpenCV, implementing neural network architectures for object detection, semantic segmentation and visual odometry in robotic perception systems
- Engineered a comprehensive course framework demonstrating integration of camera-based object detection model trained with 200 images to achieve accuracy of 95% in positive identification using deep neural networks, showcasing their crucial role in robotic perception and decision-making

Additional Professional Experience

Independent Contractor — *Agoura Hills, CA*
Mercor Intelligence — *May 2026 - Present*
- Providing domain expertise to support AI model improvement for a top AI lab

Systems Engineer — *Westlake Village, CA*
Zeolla Marble — *May 2025 - April 2026*
- Deployed and configured imaging software reducing design iteration time by 40% saving 2 hours per marble slab and streamlining AutoCAD based marble fabrication workflow
- Designed network infrastructure and camera integration pipeline for real time capture and processing systems
- Utilized CAD software, OnShape, to design of various parts to be fabricated out of carbon steel required for the project
- Collaborated with a cross-functional team, including a networking and communication specialist and a fabrication engineer, to transition from a legacy manual workflow to an automated system design.

Field Engineer II — *San Bruno, CA*
Honeywell International Inc — *Mar 2019 - Aug 2021*
- Served as the principal technical expert on HVAC systems, focusing on the integration and maintenance of commercial Fire Alarm, Security, and Access Control Systems.
- Spearheaded troubleshooting and resolution of startup issues, employing diagnostic skills to ensure optimal equipment performance.
- Developed and programmed building management controls, significantly improving building operational efficiency.
- Led instructional sessions for subcontractors on electrical and mechanical system installations, fostering a culture of sharing and collaboration.
- Mentored junior field service engineers, providing guidance and training in technical skills, customer service, and troubleshooting techniques.

Education

San Diego State University *San Diego, CA*
Master of Science, Mechanical Engineering *Aug 2021 - Dec 2024*
Emphasis: Robotics and Controls engineering

California State University, Chico *Chico, CA*
Bachelors of Science, Mechatronic Engineering *Aug 2013 - May 2018*
Automation, Robotics

Publications

Journal: IEEE Transactions on Control Systems Technology

Servos for Local Map Exploration Onboard Nonholonomic Vehicles for Extremum Seeking
Dylan James-Kavanaugh, Patrick McNamee, Qixu Wang, Zahra Nili Ahmadabadi
IEEE Transactions on Control Systems Technology pp. 1–16. 2026

Technical Skills

Languages	Python, C++, SQL
Operating System	LINUX (Ubuntu, Debian), MacOS, Windows
Robotics Software	ROS I/II, Gazebo/Mujoco, Pytorch, OpenCV, MatLab, SolidWorks/OnShape, Hugging Faces(LeRobot)
Hardware and Autonomous Planning	Sensor Fusion(LiDAR, Camera, IMU, Acoustics, Servos), RaspPi, Jetson, Motion Planning, SLAM
Development Tools	Git/GitLab/Bitbucket, Shell(Bash/Zsh), Docker, pixi, LaTeX, Microsoft Office
Soft Skills	Time Management, Teamwork, Problem-solving, Documentation, Leadership, Mentorship